

PRD_18

Connected Work Graph, Multi-User Collaboration, Document Intelligence and Cross-Module Data Flow

Construction OS

Product and engineering specification for connecting every user, company, project, client, document, task, approval and financial record through one traceable operating model.



Primary product thesis

Modules own business truth. Relationships control participation. Projects provide context. Documents provide evidence. Approvals provide authority. Events move information. Audit preserves history. AI detects missing connections without bypassing human control.

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Document Control

Field	Value
Document	PRD_18 - Connected Work Graph, Multi-User Collaboration, Document Intelligence and Cross-Module Data Flow
Product	Construction OS
Status	Discussion Draft v1.0
Purpose	Define the unified relationship, collaboration, document, event and data-access model that connects all platform modules and all internal users while preserving company, project, role and confidentiality boundaries.
Source basis	Product decisions from PRD_1 through PRD_16, repository audits of the uploaded nesto-main archives, and decisions discussed through 03 August 2026.
Precedence	This PRD extends earlier requirements. It does not replace authoritative module ownership, Company and Group governance, Platform Control Plane isolation, or existing security decisions unless explicitly stated.
Numbering note	PRD_17 remains reserved for Finance Spreadsheet Workspace, Reconciliation and AI Validation.

Core rule

Every business fact has one authoritative owner, but may be connected to many permitted users, companies, projects, clients, tasks, documents, approvals and downstream views. No module may create a second editable copy of the same fact.

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Executive Product Direction

Construction OS should not behave as a collection of disconnected departmental pages. It should operate as a connected work system in which every object can be understood through its business ownership, participants, context, evidence, decisions and history.



Figure 1. The Project Context connects work without becoming the owner of every underlying fact.

The platform must always answer seven questions

- Who created or changed this record?
- Under which Company, Group membership, role and position did the person act?
- Which Project, Client, Contractor, Contract, Unit, Department or business process provides context?
- Who is responsible now, who contributes, who reviews and who has approval authority?
- What changed, why did it change and which evidence supports it?
- Which other records or departments were affected?
- What is the user permitted to see or do in the current context?

Product outcome

The user should be able to enter from any legitimate starting point - a Project, Client, Employee, Invoice, Contract, Candidate, Drawing or Task - and reach every related record for which access is authorized. The journey must preserve one source of truth, avoid duplicate data entry and explain every relationship.

Differentiating promise

Construction OS does not merely store files and records. It preserves the complete operational story: origin, responsibility, conversation, revision, decision, approval, evidence, impact and outcome.

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Scope and Non-Goals

In scope

- Universal user, company, group, department, position and project relationship model.
- Multiple colleagues with the same user type working together on the same or different Projects.
- Record-specific participation roles such as Owner, Lead, Contributor, Reviewer, Approver, Consultant, Watcher and Delegate.
- Cross-module data movement through durable events and permission-safe projections.
- Client, contractor, candidate and other external-party relationships with internal users and records.
- Universal comments, mentions, tasks, approvals, decisions, notifications, timelines and search.
- Document provenance, immutable revisions, approval requests and Blue Approval Tickets.
- Creative collaboration, handoff, expertise and AI-exception features.
- Repository changes required to support the target architecture.

Not in scope

- Final detailed financial spreadsheet design reserved for PRD_17.
- Full Client Portal functionality. The current Client remains an internal company record unless a later PRD introduces portal access.
- Automatic AI approval, payment posting, contract execution, salary modification or document finalization.
- Replacing legal Company boundaries with a merged Group ledger or unrestricted Group access.
- Defining every department-specific field already covered by earlier module PRDs.

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Foundational Design Principles

#	Principle	Requirement
1	One authoritative owner	Each business fact is editable only in its owning module. Other pages display linked projections.
2	Relationship before role	A job title grants capability; the record relationship and active scope determine responsibility and action authority.
3	Explicit active context	Every request must use an explicit active Company, role and membership. Never select the first Company membership implicitly.
4	Company is the legal boundary	Group coordination does not merge Company ownership, ledgers, contracts, HR files or document authority.
5	Projects connect, not absorb	A Project provides shared context but does not own every contract, invoice, salary or employee record.
6	Evidence is immutable	Official documents, approved revisions, decisions and audit events must remain historically traceable.
7	Events move information	Modules react to durable domain events rather than directly editing each other's private tables.

#	Principle	Requirement
8	Deny by default	Sensitive HR, Finance, Legal and document data requires explicit permission and relationship scope.
9	Human authority remains visible	AI may detect, summarize and recommend, but official changes require an authorized human decision.
10	Explain access and change	Users should be able to see why they have access, why a record changed and which source owns the shown value.

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Connected Work Graph Domain Model

The Connected Work Graph is a logical relationship layer, not a replacement database. It connects authoritative records using typed edges. Each edge has scope, permissions, dates, source and audit history.

Primary node types

Node	Examples	Authoritative owner
Identity	Person, login, authentication factors	Identity service
Organization	Tenant, Business Group, Company, JV, Department	Organization and governance
Work context	Project, building, block, floor, unit, zone, phase	Projects / relevant domain
Party	Client, co-buyer, contractor, supplier, candidate	Owning business module
Commercial	Contract, variation, invoice, payment, budget, purchase order	Legal / Finance / Procurement
Collaboration	Task, comment, mention, approval, decision, meeting action	Shared collaboration engines
Evidence	Document, revision, attachment, inspection proof, payment proof	Document service / domain owner
Operational	Inspection, incident, stock movement, asset assignment	QA/QC, HSE, Stock, Assets
Employment	Employee, salary history, leave, performance, certification	HR

Typed relationship examples

Relationship	Meaning
MEMBER_OF	A user belongs to a Company, Group or Department under a dated membership.
ASSIGNED_TO	A user participates in a Project, task, document package or case.
OWNS / LEADS / CONTRIBUTES / REVIEWS / APPROVES	Responsibility and workflow authority for a specific record or scope.
REPRESENTS	A person represents a Client, Contractor, Supplier or Company.
RELATES_TO	A document, task, decision or event relates to another business record.
SUPERSEDES	A document revision, contract version or decision replaces a previous version without deleting history.
DERIVED_FROM	A dashboard value or projection identifies its authoritative source.
BLOCKS / DEPENDS_ON	Work dependency between tasks, approvals, documents or procurement items.
SHARED_WITH	Explicit cross-company, Group, JV or external access grant.

Relationship metadata

- Effective start and optional end date.
- Owing Company and active Company context.
- Project, department, building, unit, package or other scope.
- Permission level and confidentiality restrictions.
- Created by, reason, source workflow and timestamps.
- Whether the relationship is inherited, delegated, explicit or temporary.
- Revocation history and active-session invalidation when required.

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Identity, Company, Group and Active Context

One person, many controlled memberships

A person may work for a parent Company, support a child Company, participate in a JV Project, hold different roles in different Companies and contribute to multiple Projects. These relationships must not be compressed into one tenant role.

Layer	Required record	Purpose
Identity	User	Stable person and authentication identity.
Platform staff	PlatformStaffMembership	Separates Platform Admin authority from Company memberships.
Company	CompanyMembership	Company-specific role, status, department, position and superior.
Group	GroupMembership	Optional Group-level coordination permissions.
Project	ProjectMembership	Project-specific responsibility, discipline and scope.
Record	RecordParticipant	Owner, lead, contributor, reviewer, approver, watcher or delegate.

Layer	Required record	Purpose
Session	ActiveContext	Chosen Company, role, membership, locale and security context.

Historical actor snapshot

Every material action must store both a stable user reference and an immutable actor snapshot containing the name, position, department, active role, employing Company, active Company and Group context used at the time. A later job or Company change must not rewrite history.

Active-context requirements

- Normal Company users select or receive an explicit active Company context.
- Multi-role users select an active role. Dashboards, navigation and actions reflect that role.
- Project pages may provide a Project context but cannot silently change the user's legal Company scope.
- Cross-company access must show both the record-owning Company and the user's employing or acting Company.
- Permission changes, suspension or scope removal must invalidate affected sessions and background actions immediately.
- Platform Admin and Super Platform Admin log in through the normal login page but are redirected to the separate Platform Control Plane based on platform membership.

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User Participation and Multi-Colleague Collaboration

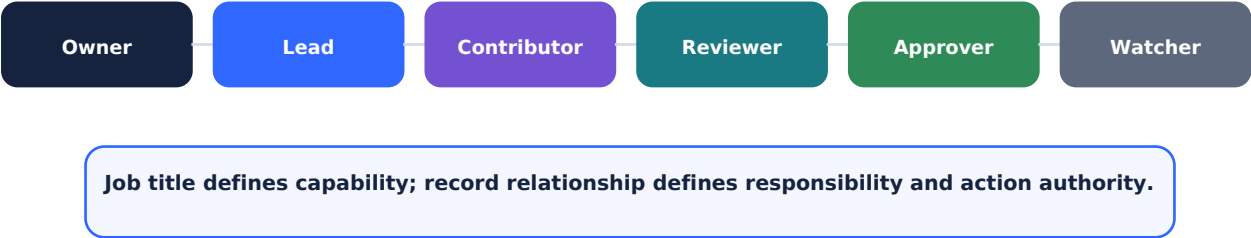


Figure 2. Responsibility is attached to a record and scope, not inferred only from a job title.

Universal participant roles

Participant	Authority	Typical use
Owner	Accountable for outcome and assignment	Candidate record, drawing package, invoice case, task, meeting action
Lead	Coordinates work and contributors	Project discipline, Finance workstream, recruitment process
Contributor	Creates work and evidence	Architect, Finance preparer, HR recruiter
Reviewer	Evaluates quality and may request changes	Peer architect, Finance reviewer, HR panel member
Approver	Issues formal approval within delegated authority	Design director, Finance approver, Company Owner
Consultant	Provides advice without ownership	Legal consulted on payment terms
Watcher	Receives selected updates without edit authority	Project Manager observing design package
Delegate	Temporarily acts under defined limits	Absence coverage
External participant	Receives restricted record-specific access	Consultant or contractor when later enabled

Same-role colleagues

The platform must allow many users with the same Company role to work together. A Project may contain ten Architects and five Finance users, but each user must have a distinct relationship, scope and current responsibility.

Example: ten Architects

Assignment dimension	Examples
Discipline	Architecture, interiors, facade, landscape, BIM coordination
Physical scope	Building, block, floor, unit type, zone
Deliverable scope	Drawing package, model, specification, revision
Workflow position	Lead, contributor, peer reviewer, approver
Phase	Concept, permit, tender, construction, as-built

Example: five Finance users

Finance participant	Project A	Project B
Finance User 1	Finance Lead	No access
Finance User 2	Invoice Preparer	Payment Reviewer
Finance User 3	Payment Approver	Budget Controller
Finance User 4	Read-only Analyst	Finance Lead
Finance User 5	No access	Invoice Preparer

Separation of duties

- The creator of a payment, salary change, invoice or approval-sensitive record may be prevented from approving the same record.

- Approval authority is checked against active role, Company, Project, amount threshold, document type and delegation.
- The system must support two-person review where required without applying that burden to every low-risk action.
- Every reassignment or delegation retains the original owner, reason and timeline.

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Project Team and Responsibility Architecture

Projects are the principal operational collaboration context. Project membership answers who may participate; record participation answers what each person is responsible for.

Project membership fields

- User and Company membership used for participation.
- Project role and discipline.
- Physical scope: building, block, floor, unit, zone or package.
- Functional scope: Finance, Legal, Architecture, Procurement, QA/QC, HSE or management.
- Permission level and sensitive-section grants.
- Start date, end date, workload allocation and availability.
- Direct lead, reporting line and substitute.
- Participation source: direct assignment, Group role, inherited team or explicit share.

Responsibility map

Every Project should display a Responsibility Map showing active departments, leads, contributors, reviewers, approvers, blocked handoffs and cross-company participation. It must be generated from live assignments rather than maintained as a decorative organization chart.

Project view	What it should answer
People by department	Who is currently involved and under which Company.
Deliverable ownership	Who owns each package and who can approve it.
Handoff status	Which department is waiting on another and for how long.
Coverage	Who acts when the responsible person is absent.
Access explanation	Why each user can see or change a Project section.
Workload	How many open responsibilities each colleague holds across Projects.

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Communication Model

Three communication levels

Level	Purpose	Governance
Record conversation	Discuss a specific Client, Candidate, Invoice, Contract, Drawing, Task or Inspection.	Preferred location for business-relevant communication; fully linked and auditable.
Project room	Coordinate cross-department work, announcements, blockers and key decisions.	Sensitive details remain hidden; messages may link governed records.
Department workspace	Allow same-type colleagues to collaborate, ask advice and peer review.	Department scope and confidentiality apply.

Message conversion

Any authorized comment or message should be convertible into a governed record while preserving the original context.

- Task or department deliverable.
- Approval request.
- Decision record.
- Risk, issue or blocker.
- RFI or clarification request.
- Meeting action.
- Document revision request.
- Access request.
- Finance review or reconciliation exception.

Mentions and colleague consultation

- A mention grants no new access by itself. The system verifies access before notifying or exposing content.
- A Smart Consultation request may provide a limited context package without granting the colleague full Project access.
- Private chat cannot be used as formal document approval, payment confirmation or contract decision.
- Important conclusions must be promoted to a Decision, Approval or Task record.
- The platform should show whether a colleague is already working on the same record to prevent duplicated effort.

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Module-to-Module Connection Specification

The following map defines the authoritative responsibility of each module and the controlled outputs consumed elsewhere.

Module	Owns	Publishes / connects to
Identity, Company and Group	Users, authentication, memberships, departments, positions, managers, active context	All authorization, actor attribution and Company/Group scope

Module	Owns	Publishes / connects to
Projects	Project identity, structure, milestones, membership, operational context	Tasks, documents, clients, contracts, budgets, teams, inspections and reporting
Tasks	Responsibility, status, deadlines, deliverables, blockers, review and closure	All source records and user Work Inboxes
Documents	File content, revision, checksum, classification, approval state, retention	Every module requiring official evidence
Clients and Sales	Client identity, contacts, representatives, purchased units, sales relationship	Contracts, Finance schedules, design requests, tasks and communications
Legal and Contracts	Contract identity, parties, terms, versions, signatures, claims and variations	Finance obligations, Project status, Client and Contractor profiles
Finance	Budgets, commitments, invoices, payments, allocations, balances, payroll projections	Projects, Client/Contractor summaries, reporting and exception workflows
HR and Recruitment	Candidates, interviews, employees, salary history, leave, performance, certifications	Finance salary values, Calendar availability, onboarding Tasks and account setup
Architecture and Design	Design packages, drawing metadata, technical review and revision requests	Documents, Client change tasks, Procurement specifications, Project milestones
Procurement	Requests, suppliers, quotations, approvals, purchase orders and deliveries	Finance commitments, Stock receipts, Project cost and schedule
Stock and Assets	Inventory, custody, movements and assignments	Projects, employees, procurement and Finance reconciliation
QA/QC and HSE	Inspections, incidents, permits, non-conformities and closure evidence	Tasks, Project risks, contractor performance and Documents
Meetings and Calendar	Meeting records, attendance, agenda, minutes and user calendar projections	Decisions, Tasks, deadlines and availability
Reporting, Search and AI	Permission-safe read models and analysis	Dashboards, search, alerts, summaries and recommendations
Platform Control Plane	Tenant lifecycle, platform staff, features, health, recovery and authorized support	Technical metadata only unless support access is explicitly granted

HR and Finance connection

HR remains the source of truth for approved salary history. Each active salary record contains currency, gross salary, net salary, effective dates and history. Finance consumes the approved gross and net values for payroll costs, forecasts and reporting without maintaining a duplicate editable salary value.

Legal and Finance connection

Contract approval must create or update Finance obligations and schedules through a domain event. A contract value is not itself an actual payment. The data model must distinguish obligation, installment schedule, invoice, payment, allocation, balance and reconciliation status.

Procurement, Stock and Finance chain

Controlled operational chain

Approved request -> quotation decision -> purchase order -> delivery -> goods receipt -> stock movement or direct Project receipt -> invoice matching -> payment -> reconciliation.

QA/QC and task connection

A failed inspection may block or reopen a related task through an event. QA/QC does not directly rewrite private task tables. Closure requires correction evidence, reinspection and an authorized decision.

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Client and External-Party Relationship Model

Client as a connected party

A Client may be an individual, company or other legal entity. Multiple co-buyers remain separate Client records connected to the same Unit and Contract. Client identity must not be duplicated for every transaction.

Client relationship	Connected records
Identity and contacts	People, representatives, communication preferences and consent
Commercial	Opportunity, reservation, purchased unit, contract, variation
Financial	Installment schedule, invoice, actual payment, allocation and balance projections
Design	Approved plans, requested modifications, selected materials and change tasks
Project	Unit, building, Project progress and permitted milestones
Communication	Comments, meetings, calls, decisions and action items
Documents	Identity evidence, contracts, plans, approvals and payment proof

Internal access

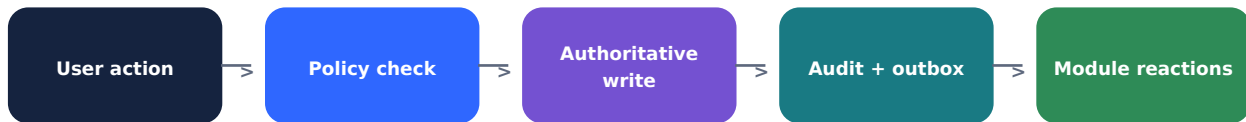
- Sales sees identity, contact, units, contract status and permitted communications.
- Finance sees financial values, schedules, payments, allocations and balances.
- Legal sees contracts, parties, legal documents and variations.
- Architecture sees unit plans and approved modification requests.
- Project Management sees permitted progress and delivery context.
- Company Admin and Owner see complete information subject to protected HR and platform boundaries.
- Every section displays its authoritative source and last authoritative update.

Contractor, supplier and candidate relationships

The same connected-party pattern should be reused for Contractors, Suppliers and Candidates while preserving domain-specific privacy. A Candidate is not visible like a Supplier; an Employee salary is not exposed because a Project Manager can see the Employee on the Project.

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Data Movement and Domain Events



Every downstream reaction carries tenant, company, actor, role, source record, correlation and confidentiality context.

Figure 3. Required path for material cross-module changes.

Context envelope

Field	Requirement
tenantId	Identifies the isolated business ecosystem.
businessGroupId	Optional Group coordination context.
owningCompanyId	Legal owner of the authoritative record.
activeCompanyId	Company context under which the actor performed the action.
projectId	Optional Project context.
actorUserId	Stable user identity.
actorMembershipId	Exact Company, Group or Platform membership used.
actorRole and position snapshot	Historical authority and organizational context.
sourceModule and sourceRecord	Authoritative origin.
eventType and schemaVersion	Stable event contract.
correlationId and causationId	Links a business process across modules and retries.
confidentiality	Controls downstream projections, notifications and AI use.
occurredAt	Server timestamp.

Event rules

- Write the authoritative record and outbox event in the same database transaction.
- Process events asynchronously with retry and dead-letter handling where appropriate.
- Consumers must be idempotent and record processed event IDs.
- Events must not contain unrestricted sensitive payloads. Consumers fetch permitted details from the owner.
- Every projection retains its source record and version.
- A failed downstream reaction must not silently roll back an already valid authoritative business decision.
- User-facing status should show pending synchronization where eventual consistency is material.

Illustrative events

Event	Typical consumers
CandidateInterviewScheduled	Calendar, notifications, recruitment collision detector
EmployeeHired	Identity onboarding, Assets, HR tasks, Finance preparation
SalaryRecordApproved	Finance payroll projection and reporting
ContractApproved	Finance obligation schedule, Project, Client/Contractor views
InvoiceConfirmed	Project financial projection, Contractor/Client profile, reconciliation
PaymentAllocated	Balances, reports, Client/Contractor summaries
DocumentRevisionSubmitted	Reviewers, approvers, linked tasks
DocumentRevisionApproved	Project, downstream packages, Blue Ticket, decision timeline
InspectionFailed	Corrective task, Project risk, contractor performance
ProjectMemberAssigned	Work Inbox, Responsibility Map, access projections

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HIGHLIGHTED REQUIREMENT: Document Passport

Non-negotiable document rule

Every online document must have a permanent Document Passport. The file, its origin, uploader context, revisions, comments, access, approvals and relationships must remain traceable for the complete retention period.

Document Passport fields

Category	Required information
Identity	Permanent document ID, title, type, classification and owning module
Organization	Owning Company, originating Company, parent Group and JV context where applicable
Uploader snapshot	Name, position, department, active role, employing Company, active Company and membership used
Time and integrity	Server upload time, file type, size, checksum, storage version and malware-scan status
Business context	Project, Client, Contractor, Contract, Unit, Task, Meeting, Invoice or other linked records
Revision	Revision code, previous revision, change summary, reason, affected pages and impact classification
Workflow	Draft/review/approval status, reviewers, approvers, comments, decisions and Blue Ticket
Security	Confidentiality, access policy, share grants, download policy and watermark rules
Retention	Retention category, legal hold, archive state and scheduled review
Usage history	Views, downloads, exports and official-reference use according to policy

Immutable origin

The original uploader metadata must be preserved as a historical snapshot. If the uploader later changes role, position, department or Company, the document continues to show the context under which it was uploaded. The current user profile may be available separately but must not rewrite the original evidence.

Group and parent-company visibility

When a user employed by a parent Company uploads a document for a child Company Project, the Document Passport must clearly show both the uploader's employing Company and the document-owning Company. Cross-company origin must never be hidden behind a generic tenant label.

Document storage requirements

- Use durable object storage rather than metadata-only file records.
- Generate immutable object versions and cryptographic checksums.
- Malware-scan uploads before making them available.
- Use signed, time-limited download URLs and policy checks.
- Prevent an approved revision from being replaced in place.
- Keep audit and file retention independent from normal record editing.
- Provide watermarking and superseded-document warnings where required.

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Revision, Approval and Blue Ticket Workflow



A new revision never inherits approval. The previously approved revision remains immutable and traceable.

Figure 4. Document revision lifecycle.

Revision submission

Uploading another revision requires a mandatory revision comment. The user must explain what changed and why. The platform should support affected pages, change category and expected cost, schedule, safety or compliance impact.

- Previous file and metadata remain immutable.
- A new revision receives a new version record and checksum.
- The uploader may compare the new revision with the previous approved or latest revision.
- Related tasks and stakeholders are notified according to their relationships.
- Comments unresolved on the previous revision may be carried as references, not silently closed.

Approval request

The uploader may tag a superior in a revision comment. The platform converts the tag into a formal Approval Request only after confirming that the selected person has approval authority for the Company, Project, document type, discipline and value or risk threshold.

- A comment or private message is not an approval.
- The approver sees the exact revision, comparison, unresolved comments and impact preview.
- The decision is Approve, Request Changes or Reject, with a mandatory comment where policy requires.
- Reauthentication may be required for high-impact approvals.
- Approval is bound to the exact revision checksum.

Blue Approval Ticket

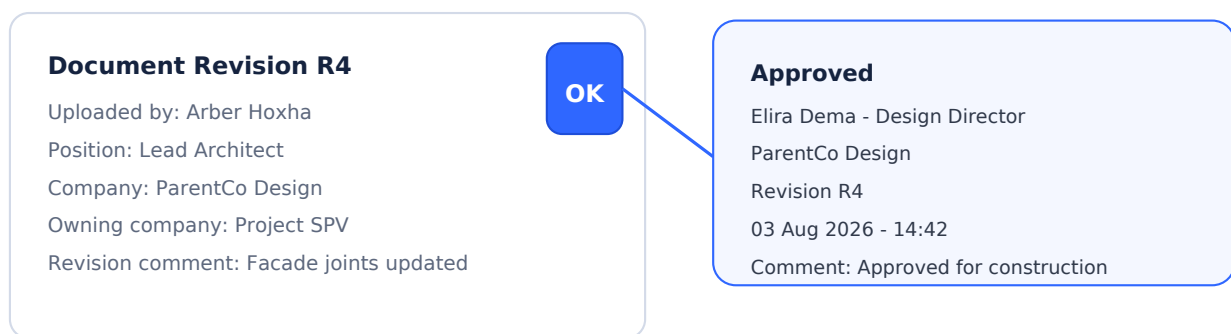


Figure 5. Blue Ticket behavior and approval bubble.

When the exact revision is approved, a blue ticket appears beside the document. Hovering or clicking the ticket shows the approver name, position, department, Company, approval date and time, revision, approval comment, authority and digital signature where used.

Revision rule

When Revision R5 is uploaded after approved Revision R4, R4 keeps its Blue Ticket. R5 starts unapproved. Approval is never inherited automatically.

Additional document intelligence

- Side-by-side revision comparison and visual redlines.
- Changed-page detection and summary.
- Approval-chain visualization.
- Linked-task and downstream-impact status.
- Expiry and scheduled-review reminders.
- Origin-Company ribbon for Group and JV documents.
- Warning when a user opens or downloads a superseded revision.
- Ability to mark a revision as Final for a defined purpose: tender, permit, construction, client approval or as-built.

14

Work Inbox, Timeline, Search and Feeds

Personal Work Inbox

Every user should receive one generated Work Inbox based on active relationships and workflow state, not only a static role dashboard.

Queue	Meaning
Needs My Action	Assigned tasks, requested reviews and pending decisions.
Waiting for My Approval	Formal approvals within current authority.
Mentioned / Consulted	Comments and consultation requests the user may access.
Submitted by My Team	Items from direct reports or led workstreams.
Waiting on Another Department	Handoffs currently blocked elsewhere.
At Risk	Overdue, missing evidence, expiring or policy-conflicting records.
Returned for Correction	Rejected or changes-requested work.
Information Only	Watched updates that require no action.

Universal activity timeline

Every major record should show a permission-filtered chronological timeline of creation, assignments, comments, document revisions, approvals, decisions, financial changes, status changes and related events. Timeline items must link back to their authoritative records.

Permission-safe search

- Search users, Projects, Clients, Contractors, Candidates, Contracts, Documents, Tasks and financial references from one interface.
- Results must never reveal restricted existence through titles, counts, snippets or suggestions.
- Each result shows owning Company, Project context, record type, status and why it is relevant.
- Users may filter by Company, Group, Project, department, record relationship, date and status.
- Search should distinguish authoritative records from generated projections.
- AI summaries must use the same result-level permissions.

Colleague awareness

Record headers should show active colleagues, current editor or revision owner, assigned lead, reviewers and pending approver. This reduces duplicate work and makes collaboration visible without becoming an unrestricted employee-monitoring feature.

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Creative Product Differentiators

Feature	Product behavior and value
Who Knows This?	Find authorized colleagues with relevant prior experience on a Client, Contractor, Candidate, system, document type or problem. Reveal expertise, not restricted record details.

Feature	Product behavior and value
Work Collision Detector	Warn when colleagues may be entering the same invoice, contacting the same candidate, revising the same drawing or sourcing the same material.
Impact Preview	Before approval, show affected tasks, budgets, purchase orders, Client commitments, documents, milestones, users and departments.
Context Handoff	Transfer work with a generated package containing current state, latest evidence, decisions, open issues, deadlines and next action.
Decision Ledger	Store the question, options, chosen decision, reasoning, authority, evidence and affected records.
Project Responsibility Map	Visualize live ownership, reviews, approvals, handoffs, workload and cross-company participation.
Evidence Completeness Score	Identify missing proof, signatures, feedback, invoices, inspection evidence or approval without making the decision automatically.
Project Memory	Generate a permission-aware weekly narrative of changes, decisions, risks, costs, delays and next actions.
Smart Consultation	Ask another colleague for targeted advice using a restricted context package rather than full Project access.
AI Exception Hunter	Detect missing or inconsistent relationships such as payment without invoice or approved drawing with unresolved comments.
Why Am I Seeing This?	Explain the membership, assignment or share that grants access and the authoritative source of displayed data.
Decision Echo	After an approval, identify users and departments that need notification, acknowledgment or follow-up action.
Handoff Heatmap	Show where work repeatedly waits between departments and which process steps create delay.
Project War Room	Temporary cross-department focus room for a critical issue, containing governed records, tasks, decisions and evidence.
Consensus Pulse	For non-binding peer review, collect structured opinions and expose disagreement before the formal approver decides.
Smart Substitute	When a responsible user is unavailable, recommend an authorized substitute based on Project scope, workload and prior involvement.

Creative design principle

Do not create novelty without governance

Every creative feature must preserve source ownership, permissions, user attribution and human authority. A feature is valuable only when it reduces missing context, duplicated work, unsafe decisions or handoff delay.

16

Permissions, Privacy and Security

Authorization formula

Policy evaluation

Identity + active membership + active Company + active role + department + Project assignment + record relationship + action + sensitivity + explicit grants/restrictions + delegation + session risk.

Security rules

- Deny access by default for sensitive records.
- Perform server-side authorization for every read and write, including server actions, APIs, background jobs, exports and AI tools.
- Never rely on stale role claims inside a long-lived token for final authorization.
- Invalidate active access immediately after suspension, membership removal or permission change.
- Use immutable or append-only audit storage for material actions.
- Require MFA and shorter sessions for Platform Admin and Super Platform Admin accounts.
- Require reauthentication for destructive, recovery, high-value financial and final-approval actions.
- Prevent Platform staff from receiving Company business-data access automatically.
- Grant temporary support access only through the approved support workflow and record every action.
- Protect salary, performance, candidate, contract, payment and confidential document data through field and record-level policies.

Privacy-preserving collaboration

- A Project Manager may see that an Employee is unavailable without seeing the leave reason or medical detail.
- Who Knows This? may identify an expert without revealing restricted case details.
- Counts and AI summaries must not leak the existence of restricted records.
- Mentions and links must be access-checked before notification content is generated.
- Cross-company collaboration must display the owning Company and never silently merge legal data.

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Proposed Data Model

The exact names may change, but the following first-class entities are required to avoid role compression, duplicate facts and untraceable collaboration.

Entity	Critical fields / purpose
User	Stable identity; no tenant role compression.
CompanyMembership	userId, companyId, roleId, departmentId, positionId, managerId, status, dates.
GroupMembership	userId, groupId, group role, company scope, dates.
PlatformStaffMembership	platform role, permission package, customized permissions, Company scope, status.
ProjectMembership	userId, membershipId, projectId, discipline, scope, project role, dates, workload.

Entity	Critical fields / purpose
RecordParticipant	record type/id, user/membership, participant type, scope, authority, dates.
Delegation	grantor, delegate, action scope, record/company/project scope, dates and reason.
Comment	author snapshot, source record, visibility, mentions and conversion links.
Decision	question, options, decision, reasoning, authority, affected records and evidence.
ApprovalRequest	target record/version, requested approver, authority policy, state and decisions.
Document	logical document identity, owning Company/module, classification and retention.
DocumentRevision	immutable object version, checksum, revision code, change summary, uploader snapshot and state.
DocumentApproval	exact revision checksum, approver snapshot, decision, comment and timestamp.
DomainEvent / OutboxEvent	event envelope, schema version, payload reference and publishing state.
ProcessedEvent	consumer idempotency record.
ActivityEvent	permission-filterable timeline representation.
Notification	recipient, reason, source, priority, state and delivery channel.
AccessGrant	explicit record, Project, Company, JV, support or external share.

Financial and salary precision

- All monetary values use decimal types with explicit currency.
- Salary history stores both gross and net salary, currency, frequency, effective dates, status and audit history.
- Financial obligations, schedules, invoices, payments, allocations and balances are separate entities.
- Displayed values identify the authoritative Finance or HR source and record version.

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Event and API Contracts

Command and query separation

Commands change authoritative state after policy and validation. Queries return permission-safe views. A UI page may combine multiple query projections, but it may not write directly into multiple modules as one uncontrolled client operation.

Required command behavior

- Validate active context and action authority.
- Validate all related Company, Project and tenant identifiers server-side.
- Apply optimistic concurrency or version checks for collaborative edits.
- Write audit and outbox atomically.
- Return authoritative version and any pending downstream synchronization state.
- Use idempotency keys for retried external or high-impact requests.

Illustrative document commands

Command	Required validation
CreateDocument	Owning Company/module, uploader context, file scan and classification.
UploadDocumentRevision	Document access, mandatory change summary, immutable object version and checksum.
RequestDocumentApproval	Exact revision, eligible approver and authority policy.
ApproveDocumentRevision	Approver authority, unresolved requirements, exact checksum and reauthentication where required.
SupersedeDocumentRevision	New approved revision or authorized administrative decision.
GrantDocumentAccess	Grantor authority, recipient scope, expiration and confidentiality.

Concurrency

- Display when another user is editing the same collaborative record.
- Use field or record version checks before saving.
- Never silently overwrite another user's changes.
- For documents, collaboration occurs through comments and new revisions rather than replacing the same approved file.
- For structured records, show a conflict comparison and require an explicit merge or retry.

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UX Requirements

Universal record header

Element	Requirement
Identity	Record title, type, ID, status and owning Company.
Context	Project, Client/Contractor, Contract, Unit and source module.
Responsibility	Owner, lead, contributors, reviewer, approver and substitute.
Activity	Last authoritative update and responsible user.
Actions	Only actions authorized in the active context.
Access explanation	Why the current user can view or act.
Evidence state	Document completeness, approval and revision status.

Document viewer

- Document Passport side panel.
- Blue Ticket anchored beside approved revision.
- Revision selector with approved, superseded and draft states.
- Compare revisions and highlight changed pages.
- Comments tied to revision and optional page/area reference.
- Formal approval request action separate from normal comment.
- Linked records, tasks, decisions and downstream impact.

- Superseded warning and permitted download controls.

Project collaboration view

- Responsibility Map and Project team filters.
- Cross-department handoff board.
- Project Room with governed-record conversion.
- Current blockers, pending approvals and expiring documents.
- Company badges for parent, child, JV and external participation.
- Permission-filtered Project Memory summary.

User experience consistency

- English and Albanian localization.
- EU number and date formatting, with explicit currency display.
- Platform Default, Night Mode and role-adapted Creative User Type themes.
- Fixed status meanings and colors across themes.
- Responsive layouts, keyboard access, tooltips, skeleton states and clear empty states.
- No drag-and-drop customization that changes governed responsibility or hides mandatory work queues.

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AI Agent Boundaries

Permitted AI functions

- Summarize permitted Project, record and document history.
- Detect missing links, duplicates, anomalies and incomplete evidence.
- Compare document revisions and propose a change summary.
- Suggest responsible colleagues based on authorized expertise and workload.
- Generate draft tasks, decisions, handoff packages and meeting actions for human confirmation.
- Reconcile Finance spreadsheet rows against authoritative invoices, payments, contracts and expenses under PRD_17.
- Explain why a record is inconsistent or at risk using linked evidence.

Prohibited autonomous actions

- Approving or finalizing documents.
- Posting payments, invoices, journal entries or salary changes.
- Executing or modifying contracts.
- Changing Company ownership, memberships or permissions.
- Deleting records or bypassing retention.
- Granting access or revealing restricted information.
- Closing inspections, incidents, tasks or recruitment decisions without authorized human confirmation.

AI security

- AI retrieval uses the same server-side authorization as normal queries.
- Prompts and outputs inherit record confidentiality.
- Restricted record existence must not be inferable through AI counts or descriptions.
- Every AI recommendation links to the records and evidence used.
- Every accepted AI suggestion records the human actor and final changes.
- AI model providers must not receive unrestricted tenant data without approved governance and contractual controls.

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Repository Alignment and Migration

The uploaded repository demonstrates broad module coverage, but the target architecture requires foundational corrections before advanced collaboration and AI features can be trusted.

Observed architecture risk	Required direction
Tenant-only membership and role compression	Introduce Company, Group, Project and record participation relationships.
Implicit first-Company selection	Require explicit active Company and membership context.
Stale role data in tokens or server actions	Authorize against current server-side membership and policy state.
Direct cross-module writes	Adopt authoritative commands, outbox events and idempotent consumers.
Partial event metadata and synchronous handling	Add full context envelope, durable worker, retries and dead-letter monitoring.
Financial concepts combined incorrectly	Separate obligation, schedule, invoice, payment, allocation, balance and reconciliation.
Floating-point money	Migrate to decimal values and explicit currency.
Document metadata without complete file governance	Add object storage, immutable versions, checksum, scan and signed access.
Mutable or tenant-deletable audit	Create append-only audit and protected retention.
Platform Admin represented mainly by a flag	Implement PlatformStaffMembership, delegated permissions, Company scopes and separate control plane.
Weak authentication and temporary credentials	Forced password change, expiry, rate limits, MFA, session controls and recovery governance.
Insufficient automated tests and CI	Add authorization, isolation, workflow, event, concurrency and document-integrity tests.

Migration strategy

- Create new relationship tables without removing current Company behavior.
- Backfill Company memberships and explicit active contexts.
- Introduce policy evaluation behind existing server actions, then centralize it.
- Add outbox events alongside current writes before removing direct module coupling.
- Migrate documents to object storage and create initial revision records.
- Backfill actor snapshots and audit metadata where reliable source data exists.

- Convert monetary values to decimal with reconciliation reports before cutover.
- Enable Connected Work Graph features Project by Project or tenant by tenant behind feature flags.

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Acceptance Criteria

Area	Acceptance criterion
Multi-user collaboration	Ten Architects or five Finance users can participate in one Project with distinct scopes, responsibilities and approval authority.
Cross-project assignment	A user may lead one Project, review another and have no access to a third without duplicate accounts.
Company context	Every action identifies owning Company, active Company and exact membership.
Colleague awareness	Authorized users can see who owns, contributes, reviews and approves the current record.
Communication	A record comment can become a linked task, approval, decision or consultation without losing origin.
HR example	A second HR user can discover an existing Candidate process and contribute permitted feedback without creating a duplicate candidate.
Document passport	Every document revision exposes required provenance, Company and uploader snapshot fields.
Revision control	A new revision requires a change summary and never overwrites the previous file.
Blue Ticket	The approved revision shows a Blue Ticket with the correct approver bubble; a later revision does not inherit it.
Authoritative ownership	Client, Project and Contractor pages display Finance values from Finance rather than independent editable copies.
Event reliability	A retried event does not create duplicate tasks, payments, projections or notifications.
Permission safety	Search, timeline, notifications, counts and AI do not expose restricted records.
Audit	Material changes show actor, role, position, Company, time, reason and prior/new state.
AI governance	AI suggestions cannot finalize official actions without authorized human confirmation.
Immediate revocation	Removing a permission or Company scope blocks active sessions and background actions immediately.

Document workflow scenario

Step	Expected result
1	Architect employed by Parent Company uploads Revision R3 for a child Company Project. Passport shows both Companies and the historical actor position.
2	Uploader enters mandatory change summary and tags an eligible superior.
3	System creates a formal Approval Request linked to R3 and verifies authority.
4	Approver requests changes. R3 remains immutable and the comment appears in history.
5	Uploader creates R4 with a new checksum and explains corrections.
6	Approver approves R4. Blue Ticket appears beside R4 with the approval bubble.
7	R5 is later uploaded. R4 remains approved and immutable; R5 has no Blue Ticket.
8	Project, task and downstream users receive permission-safe updates through events.

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Implementation Sequence

Phase	Scope	Exit condition
0 - Trust foundation	Active context, policy engine, Company validation, session invalidation, Owner/Admin and Platform isolation	Cross-tenant and stale-role vulnerabilities closed.
1 - Evidence foundation	Object storage, document revisions, checksums, immutable audit, decimal money	Official evidence and Finance values are trustworthy.
2 - Shared collaboration engines	Comments, mentions, participants, tasks, approvals, decisions, notifications and timeline	All modules use common governed collaboration services.
3 - Event architecture	Outbox, workers, idempotency, context envelope, projection monitoring	Cross-module reactions are durable and traceable.
4 - Connected Work Graph UX	Responsibility Map, Work Inbox, record header, search, Project/department rooms	Users can understand people, context and dependencies.
5 - Document intelligence	Passport, comparison, Blue Ticket, impact preview and expiry	Document lifecycle operates end to end.
6 - Collaboration intelligence	Collision detection, Who Knows This?, handoff, Project Memory and consultation	Duplicate work and knowledge loss are reduced.
7 - AI exceptions	Permission-safe anomaly detection and recommendations, plus PRD_17 reconciliation	AI adds value without becoming an authority.

Recommended first vertical slice

Implement one complete workflow rather than isolated UI elements: Client design request -> Architecture task -> document revision -> superior approval -> Blue Ticket -> Project update -> Finance/Legal impact review where required. This slice tests identity context, participants, comments, tasks, documents, approvals, events, notifications and audit together.

Open Decisions

Decision	Options requiring later confirmation
Formal document approval depth	Single approver; sequential chain; parallel approvals by discipline; threshold-based combination.
Peer review consensus	Informational only; required quorum; configurable by document type.
Team availability visibility	Exact users who may see named leave versus only anonymous unavailability.
Unassigned user behavior	Read only; request participation; create personal draft; department default queue.
External collaboration	When and how Clients, contractors and consultants receive portal or record-specific access.
Intercompany Finance presentation	Gross only; eliminated only; both gross and consolidated after elimination.
Document digital signature	Internal authenticated approval only; qualified electronic signature integration; configurable.
Automatic reassignment	Recommendation only; automatic substitute; lead-confirmed reassignment.
Support co-assignment authority	Equal authority; lead/assistant model; second confirmation for high-impact support actions.

Final architecture position

Construction OS should become a traceable collaboration operating system: modules own truth, Projects connect work, users participate through explicit relationships, documents preserve evidence, approvals preserve authority, events move information, audit preserves history and AI identifies missing connections.